



Lab Task 2

Only for course Teacher					
	Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage	25%	50%	75%	100%	
Clarity					
Content Quality					
Spelling & Grammar					
Organization and Formatting					
Total obtained mark					
Comments					

Semester: Spring / Fall

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Batch: 46

Section:B

Course Code: CSE 124

Course Name: Object Oriented Programming

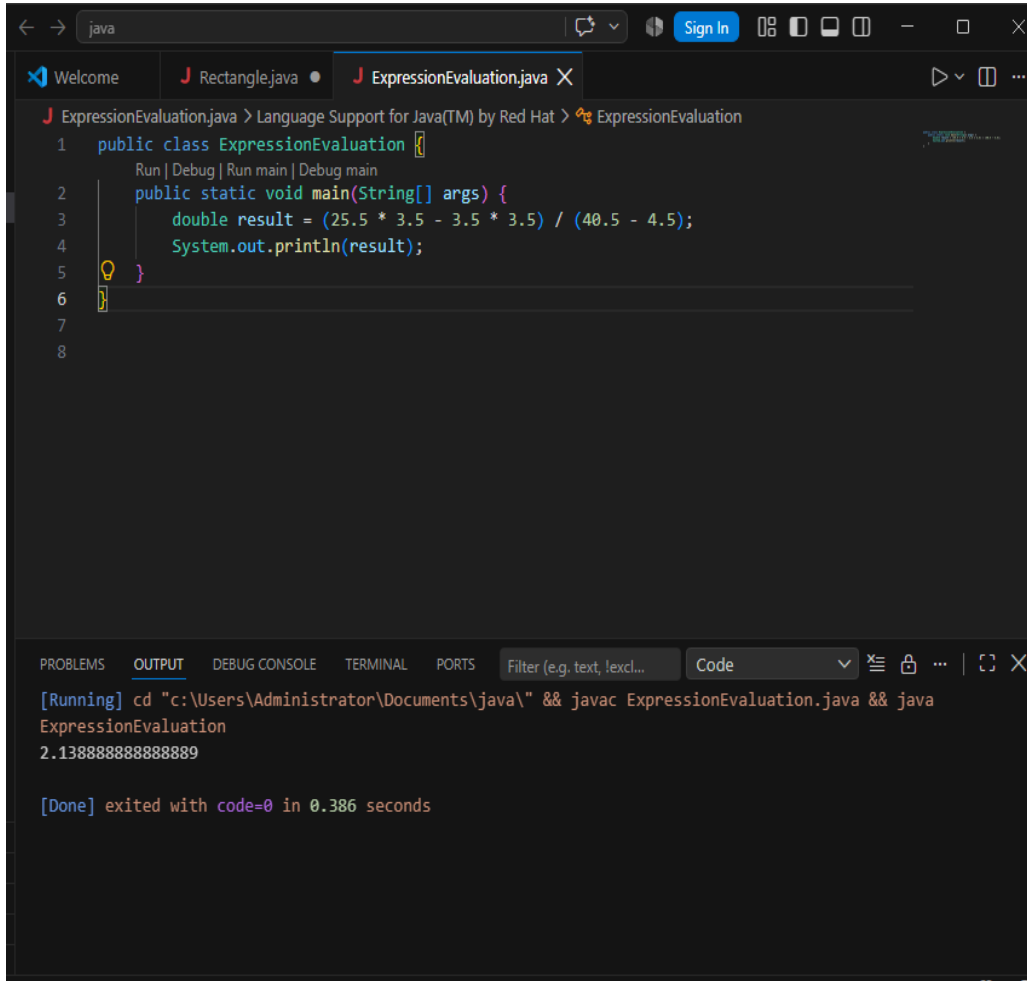
Course Teacher Name: Debabarata Mallick

Designation:Lecturer

Submission Date :13\ 08\ 2026

Experiment No : 09

Experiment Name : Expression Evaluation .



The screenshot shows an IDE window with a Java file named `ExpressionEvaluation.java`. The code defines a class `ExpressionEvaluation` with a `main` method that calculates the value of the expression $(25.5 * 3.5 - 3.5 * 3.5) / (40.5 - 4.5)$. The IDE's output console shows the command used to run the program and the resulting output, which is `2.138888888888889`.

```
public class ExpressionEvaluation {  
    public static void main(String[] args) {  
        double result = (25.5 * 3.5 - 3.5 * 3.5) / (40.5 - 4.5);  
        System.out.println(result);  
    }  
}
```

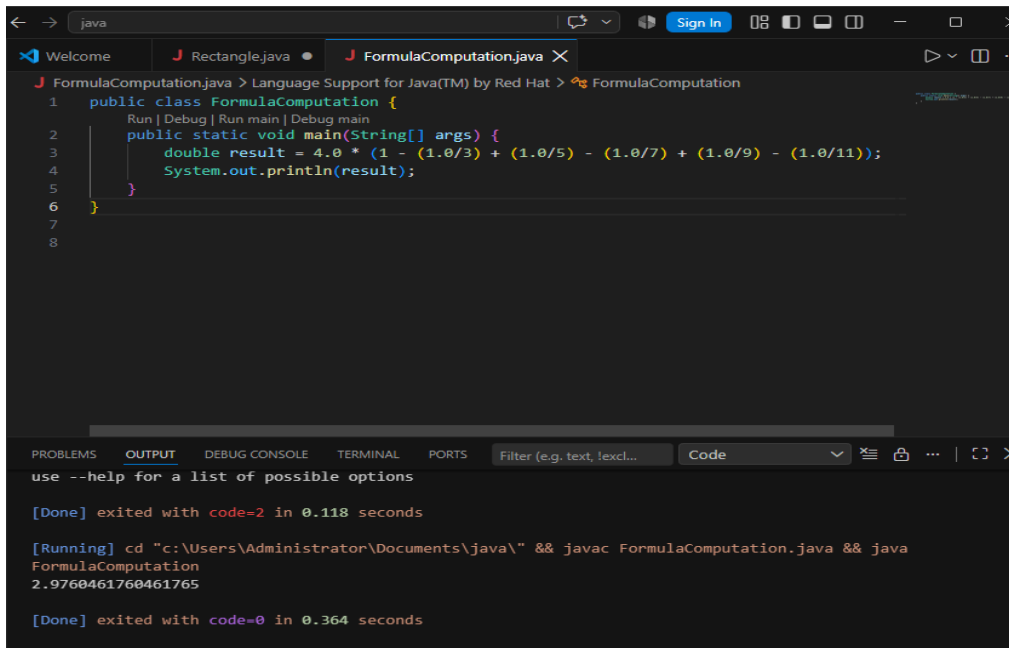
[Running] `cd "c:\Users\Administrator\Documents\java\" && javac ExpressionEvaluation.java && java ExpressionEvaluation`
`2.138888888888889`

[Done] exited with code=0 in 0.386 seconds

Discussion: This program calculates a mathematical expression with multiplication and division. I followed the given formula step by step. The final result came out to be approximately 2.14.

Experiment No : 10

Experiment Name : Formula Computation.



The screenshot shows an IDE window with a Java file named `FormulaComputation.java`. The code defines a public class `FormulaComputation` with a static `main` method. Inside the `main` method, a `double` variable `result` is calculated using the formula $4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11))$, and the result is printed to the console. The IDE's output window at the bottom shows the execution results, including the command used to run the program and the output value `2.9760461760461765`.

```
1 public class FormulaComputation {
2     Run | Debug | Run main | Debug main
3     public static void main(String[] args) {
4         double result = 4.0 * (1 - (1.0/3) + (1.0/5) - (1.0/7) + (1.0/9) - (1.0/11));
5         System.out.println(result);
6     }
7
8 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excl... Code

use --help for a list of possible options

[Done] exited with code=2 in 0.118 seconds

[Running] cd "c:\Users\Administrator\Documents\java\" && javac FormulaComputation.java && java FormulaComputation

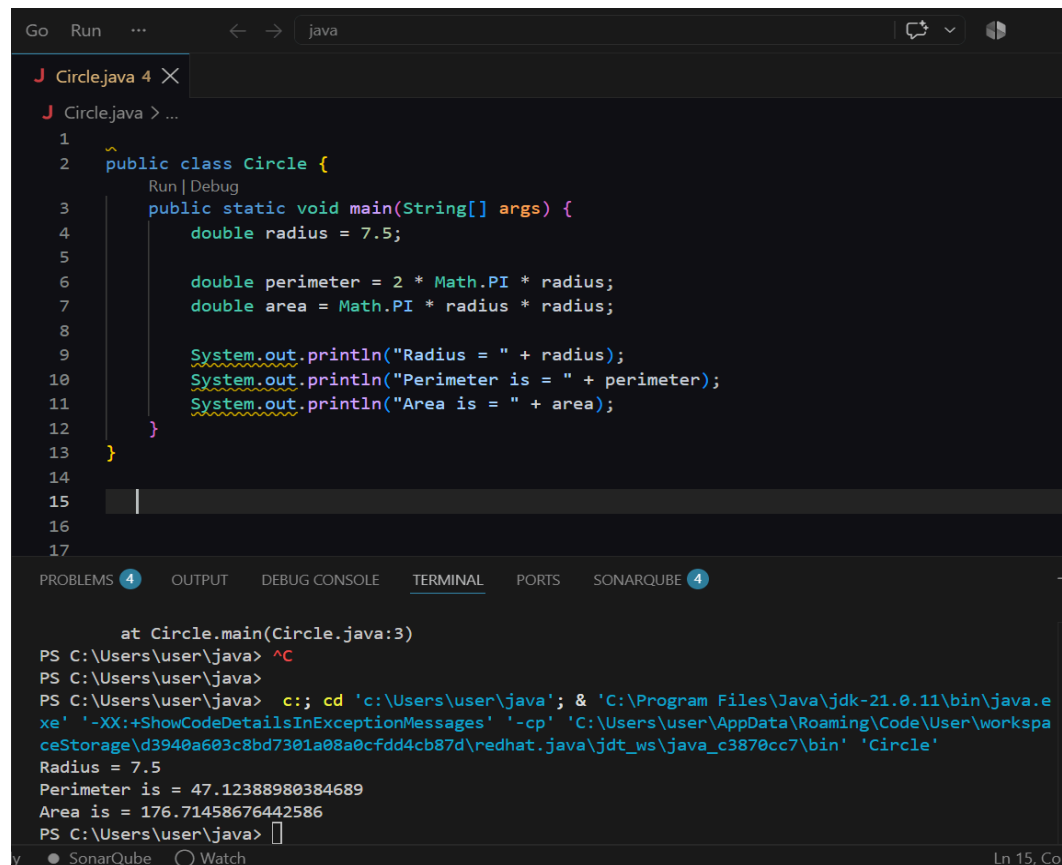
2.9760461760461765

[Done] exited with code=0 in 0.364 seconds

Discussion : This formula is used to find an approximate value of pi. I calculated the given terms using floating point numbers. The output I got is around 2.976.

Experiment No: 11

Experiment Name : Circle Area And Perimeter

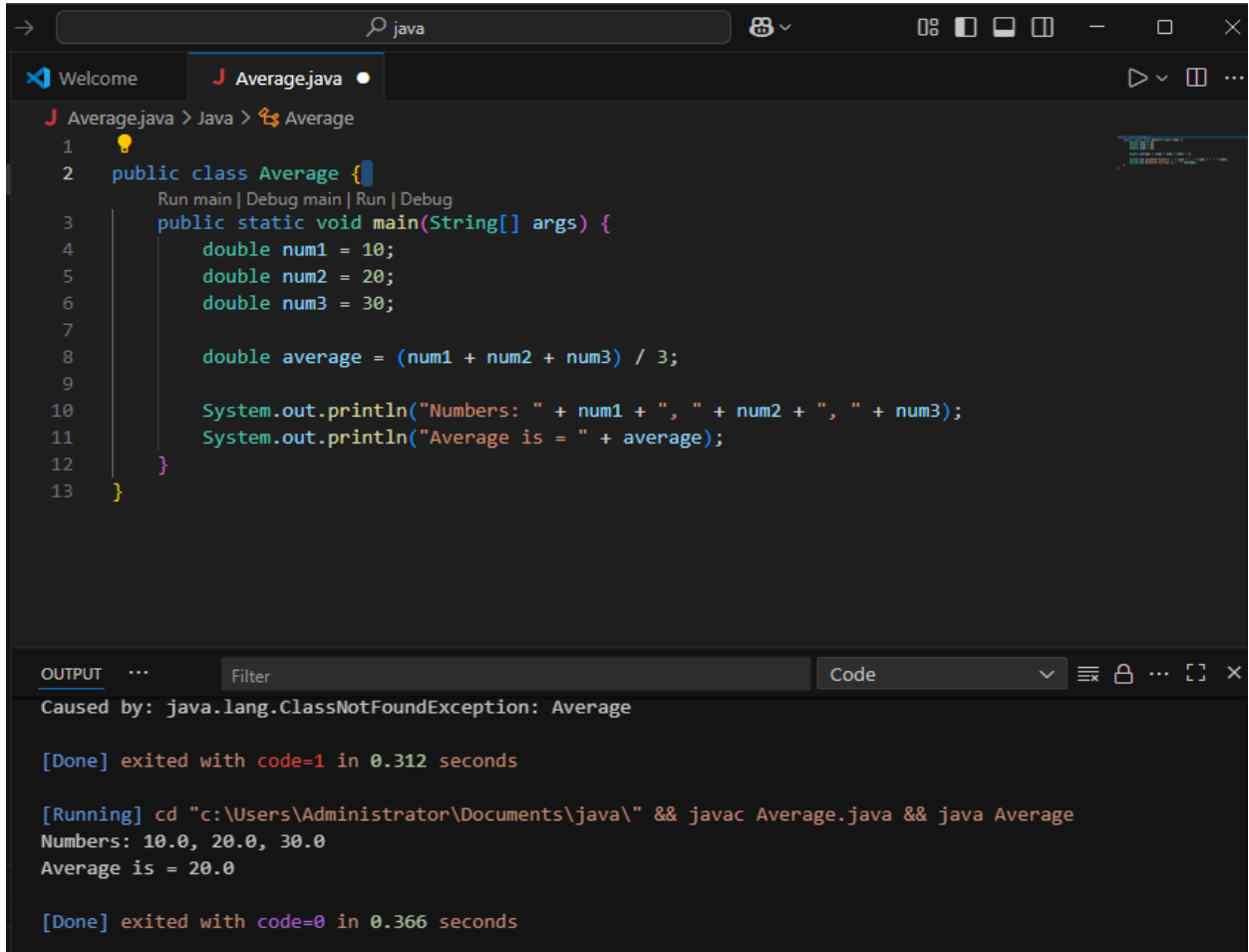


```
Go Run ... java
J Circle.java 4 X
J Circle.java > ...
1
2 public class Circle {
3     Run | Debug
4     public static void main(String[] args) {
5         double radius = 7.5;
6
7         double perimeter = 2 * Math.PI * radius;
8         double area = Math.PI * radius * radius;
9
10        System.out.println("Radius = " + radius);
11        System.out.println("Perimeter is = " + perimeter);
12        System.out.println("Area is = " + area);
13    }
14
15
16
17
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS SONARQUBE 4
    at Circle.main(Circle.java:3)
PS C:\Users\user\java> ^C
PS C:\Users\user\java>
PS C:\Users\user\java> c:: cd 'c:\Users\user\java'; & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'Circle'
Radius = 7.5
Perimeter is = 47.12388980384689
Area is = 176.71458676442586
PS C:\Users\user\java>
y ● SonarQube ○ Watch Ln 15, Col
```

Discussion: I calculated the area and perimeter of a circle using radius 7.5. Used the formulas $2\pi r$ for perimeter and πr^2 for area.

Experiment No: 12.

Experiment Name : Average Of Three Numbers



```
1 public class Average {
2     public static void main(String[] args) {
3         double num1 = 10;
4         double num2 = 20;
5         double num3 = 30;
6
7         double average = (num1 + num2 + num3) / 3;
8
9         System.out.println("Numbers: " + num1 + ", " + num2 + ", " + num3);
10        System.out.println("Average is = " + average);
11    }
12 }
13 }
```

OUTPUT

Caused by: java.lang.ClassNotFoundException: Average

[Done] exited with code=1 in 0.312 seconds

[Running] cd "c:\Users\Administrator\Documents\java\" && javac Average.java && java Average

Numbers: 10.0, 20.0, 30.0

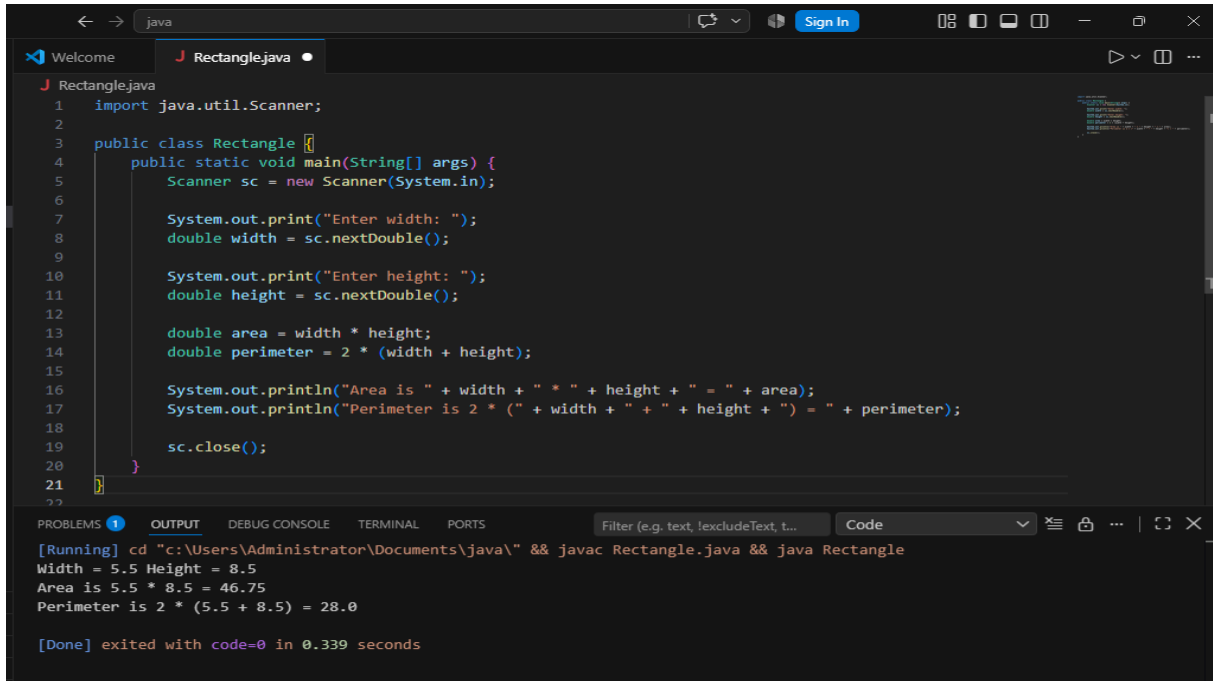
Average is = 20.0

[Done] exited with code=0 in 0.366 seconds

Discussion: I took three numbers and found their average by adding them and dividing by 3 .

Experiment No: 13.

Experiment Name : Rectangle Area And Perimeter .



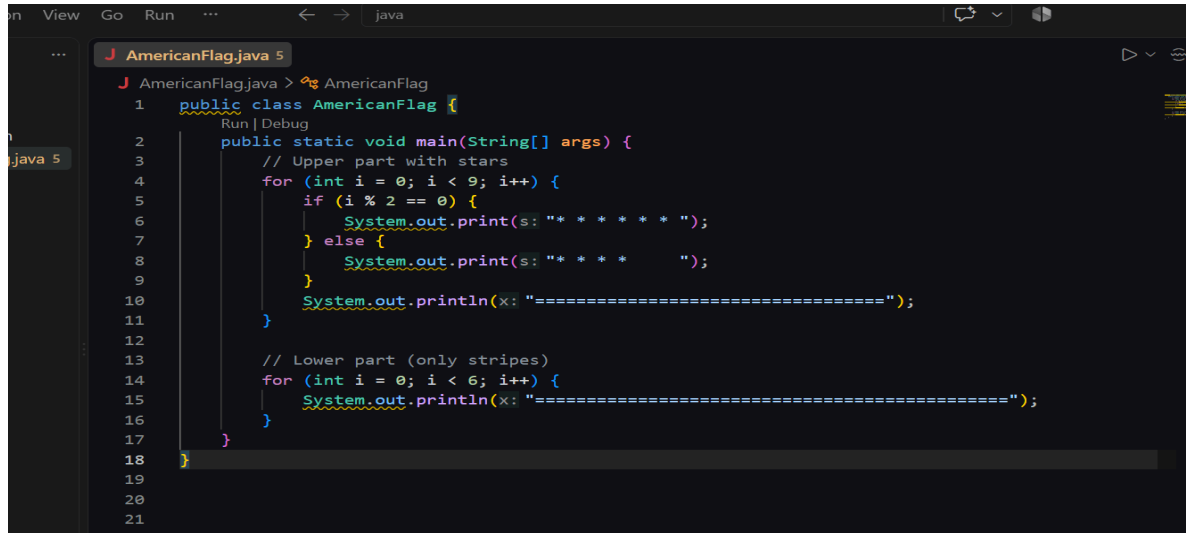
```
1 import java.util.Scanner;
2
3 public class Rectangle {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter width: ");
8         double width = sc.nextDouble();
9
10        System.out.print("Enter height: ");
11        double height = sc.nextDouble();
12
13        double area = width * height;
14        double perimeter = 2 * (width + height);
15
16        System.out.println("Area is " + width + " * " + height + " = " + area);
17        System.out.println("Perimeter is 2 * (" + width + " + " + height + ") = " + perimeter);
18
19        sc.close();
20    }
21 }
```

[Running] cd "c:\Users\Administrator\Documents\java\" && javac Rectangle.java && java Rectangle
Width = 5.5 Height = 8.5
Area is 5.5 * 8.5 = 46.75
Perimeter is 2 * (5.5 + 8.5) = 28.0
[Done] exited with code=0 in 0.339 seconds

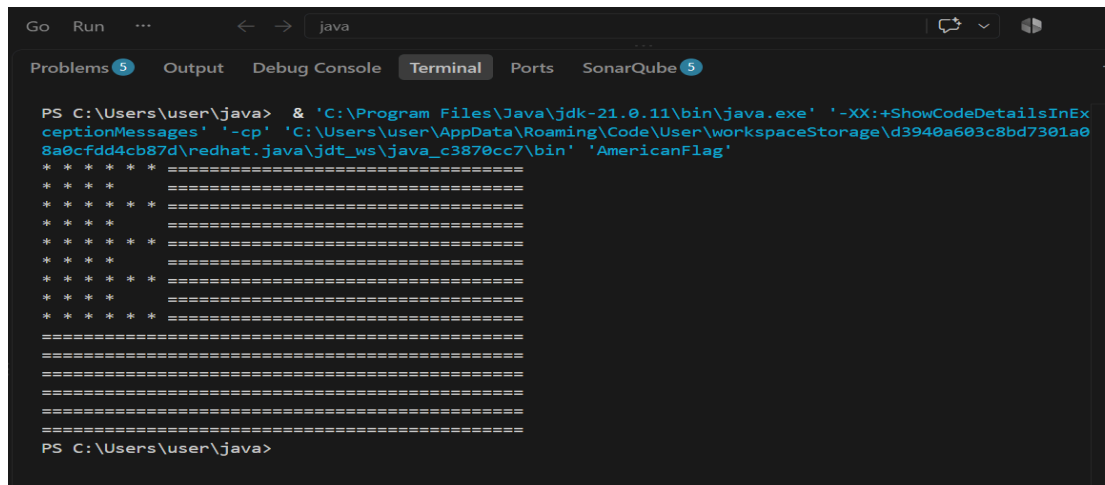
Discussion : I calculate the area and perimeter of a rectangle using width 5.5 and height 8.5 with the formulas length and width and $2(l+w)$.

Experiment No : 14

Experiment Name : American Flag Display .



```
1 public class AmericanFlag {
2     public static void main(String[] args) {
3         // Upper part with stars
4         for (int i = 0; i < 9; i++) {
5             if (i % 2 == 0) {
6                 System.out.print(s: "* * * * * ");
7             } else {
8                 System.out.print(s: "* * * * ");
9             }
10            System.out.println(x: "=====");
11        }
12
13        // Lower part (only stripes)
14        for (int i = 0; i < 6; i++) {
15            System.out.println(x: "=====");
16        }
17    }
18 }
19
20
21
```

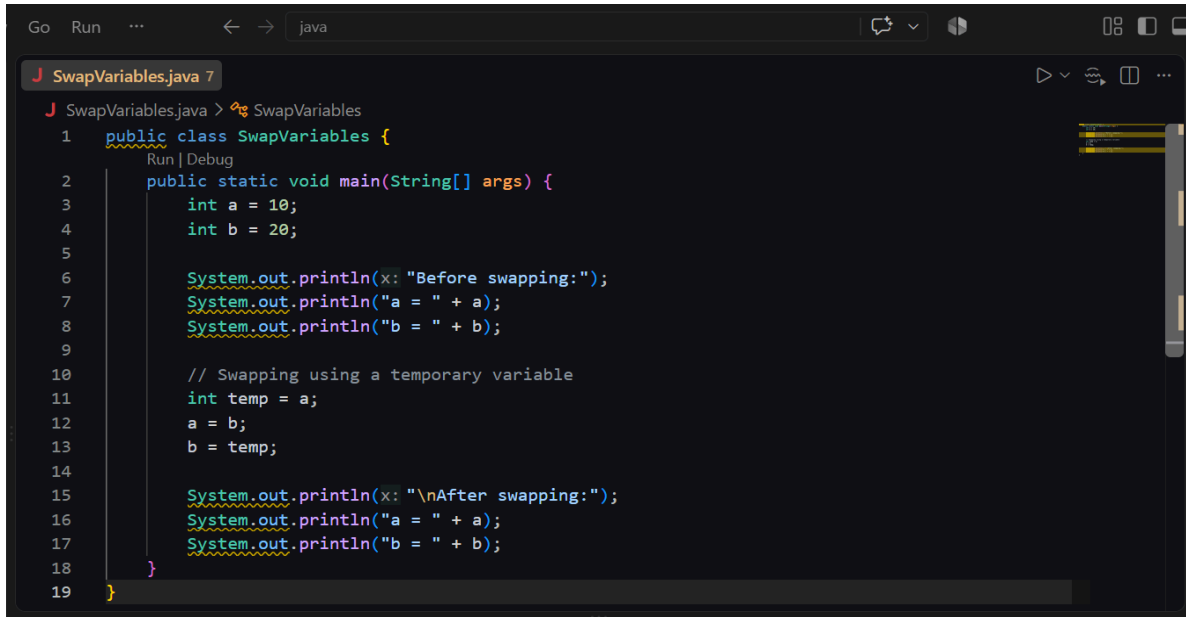


```
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'AmericanFlag'
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
* * * * * =====
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=====
PS C:\Users\user\java>
```

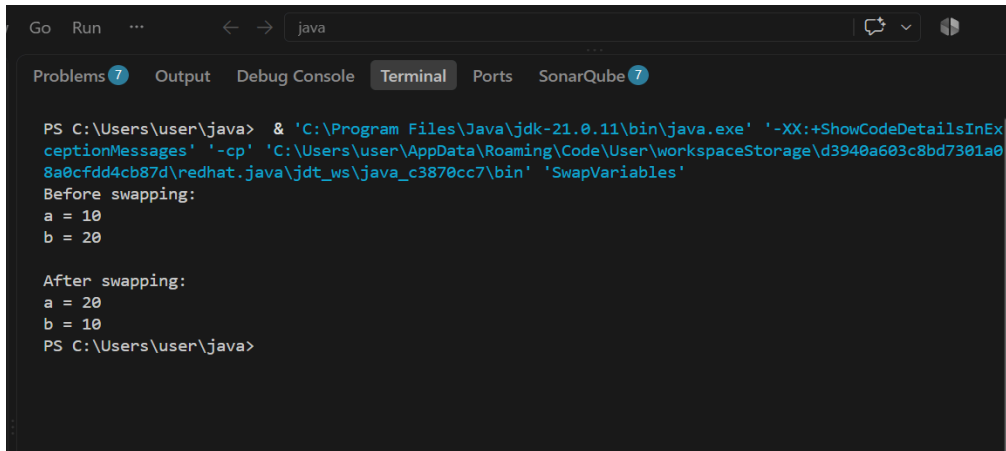
Discussion:I printed an american flag pattern using loops with stars and equal signs to make the stars.

Experiment No : 15

Experiment Name: Swap Two Variables.



```
Go Run ... < -> java
J SwapVariables.java 7
J SwapVariables.java > SwapVariables
1 public class SwapVariables {
    Run | Debug
2     public static void main(String[] args) {
3         int a = 10;
4         int b = 20;
5
6         System.out.println(x: "Before swapping:");
7         System.out.println("a = " + a);
8         System.out.println("b = " + b);
9
10        // Swapping using a temporary variable
11        int temp = a;
12        a = b;
13        b = temp;
14
15        System.out.println(x: "\nAfter swapping:");
16        System.out.println("a = " + a);
17        System.out.println("b = " + b);
18    }
19 }
```



```
Go Run ... < -> java
Problems 7 Output Debug Console Terminal Ports SonarQube 7
PS C:\Users\user\java> & 'C:\Program Files\Java\jdk-21.0.11\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\d3940a603c8bd7301a08a0cfdd4cb87d\redhat.java\jdt_ws\java_c3870cc7\bin' 'SwapVariables'
Before swapping:
a = 10
b = 20

After swapping:
a = 20
b = 10
PS C:\Users\user\java>
```

Discussion : I swapped the values of two variables using a temporary variable and printed the values before and after swapping.